GILGIT · CLIMATE REGRESSION

Predicting Gilgit’s heat extremes

How a simple linear regression connects monthly cold-night minimums to daytime maximums in historical climate records.

DATA SCIENCE STUDY · MONTHLY CLIMATE AGGREGATES

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THE QUESTION

Can one extreme predict the other?

INPUT · X

Lowest Minimum Temperature

Monthly cold-night minimums.

OUTPUT · Y

Highest Maximum Temperature

Monthly daytime maximums.

Monthly climate aggregates from Gilgit, Pakistan.

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WHY THIS SETTING

Gilgit’s climate creates a useful test.

CLIMATE CONTEXT

High-altitude valley climate

Gilgit’s high-altitude valley climate produces pronounced seasonal swings.

ANALYTIC MOTIVE

Do the extremes move together?

That variation creates a useful setting for testing how monthly temperature extremes move together.

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THE DATA

The dataset combines temperature and precipitation records

SOURCE FILE

Climate records of Gilgit.csv

MONTHLY CLIMATE AGGREGATES

The analysis works with monthly climate aggregates rather than annual summaries.

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DATA PREPARATION

Three cleaning moves make the data model-ready

01 Load the CSV

Load the CSV into a pandas DataFrame.

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02 Normalize column names

Strip leading and trailing whitespace from column names.

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03 Keep monthly rows only

Remove the **Annual** summary row so it does not bias the monthly regression.

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FEATURE DESIGN

Feature selection turns the climate question into X and y

X

Lowest Minimum Temperature in Celsius

→

Y

Highest Maximum Temperature in Celsius

The model learns whether warmer monthly lows correspond to warmer monthly highs.

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01

PART 01

Methods

From filtered records to a fitted line.

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METHOD · ORDINARY LEAST SQUARES

Simple linear regression fits the best straight line

The model uses Ordinary Least Squares through scikit-learn's **LinearRegression**.

$$y = \beta_1 x + \beta_0$$

WHAT THE LINE DOES

Minimizes the residual error

It fits the best straight line linking monthly lowest minimums to highest maximums.

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REPRODUCIBILITY

Training is a short, reproducible pipeline

01 Extract **X** and **y** from the filtered monthly DataFrame.

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02 Fit the model with `model.fit(X, y)`.

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03 Generate fitted values with `model.predict(X)`.

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04 Calculate the slope, intercept, R^2, and RMSE.

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THE FITTED MODEL

The fitted equation translates cold nights into predicted highs

PREDICTION RULE

Predicted maximum temperature (°C) = 1.2136 × lowest minimum temperature (°C) + 35.6320.

SLOPE · B_1

1.2136

The fitted multiplier on the lowest minimum temperature.

INTERCEPT · B_0

35.6320

The fitted baseline in the equation.

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